Countering Disinformation with
SOURCE
Crawling
Classification
Contextualization
Community-Building

Scalable, Open and Comprehensive Recognition of Disinformation Campaigns on the Web

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1. Overview

SOURCE At A Glance

SOURCE builds an open European infrastructure for detecting and analysing disinformation campaigns across web and social-media content.

Core idea

Collect large-scale digital traces, analyse them with AI (and if they have been created by AI), and publish potential disinformation artefacts in an open database.

Infrastructure base

Extend **OpenWebSearch.eu** and the **Open Web Index** with particular social media instead of depending on closed search and platform APIs.

Project structure

Five linked modules: provenance analysis, content analysis, dissemination, infrastructure, and operational use cases.

Expected result

Open data, open interfaces, reproducible workflows, and practical support for researchers, analysts, and fact-checking communities. Shared Tasks.

> Open for collaboration with other DESINFO Projects where SOURCE can serve as data source.

Consortium Overview

University of Passau

Coordination, crawling extension, provenance analysis, diffusion modelling, and platform integration.

UNI KASSEL VERSITÄT

University of Kassel

AI-generated content detection, fact checking, rhetoric and argumentation analysis, and evaluation.

osf open search foundation

Open Search Foundation

Community building, dissemination, open-source transfer, and stakeholder-facing activities.

lrz

LRZ

HPC backbone, federated storage, and scalable workflow orchestration.

Alliance 4Europe

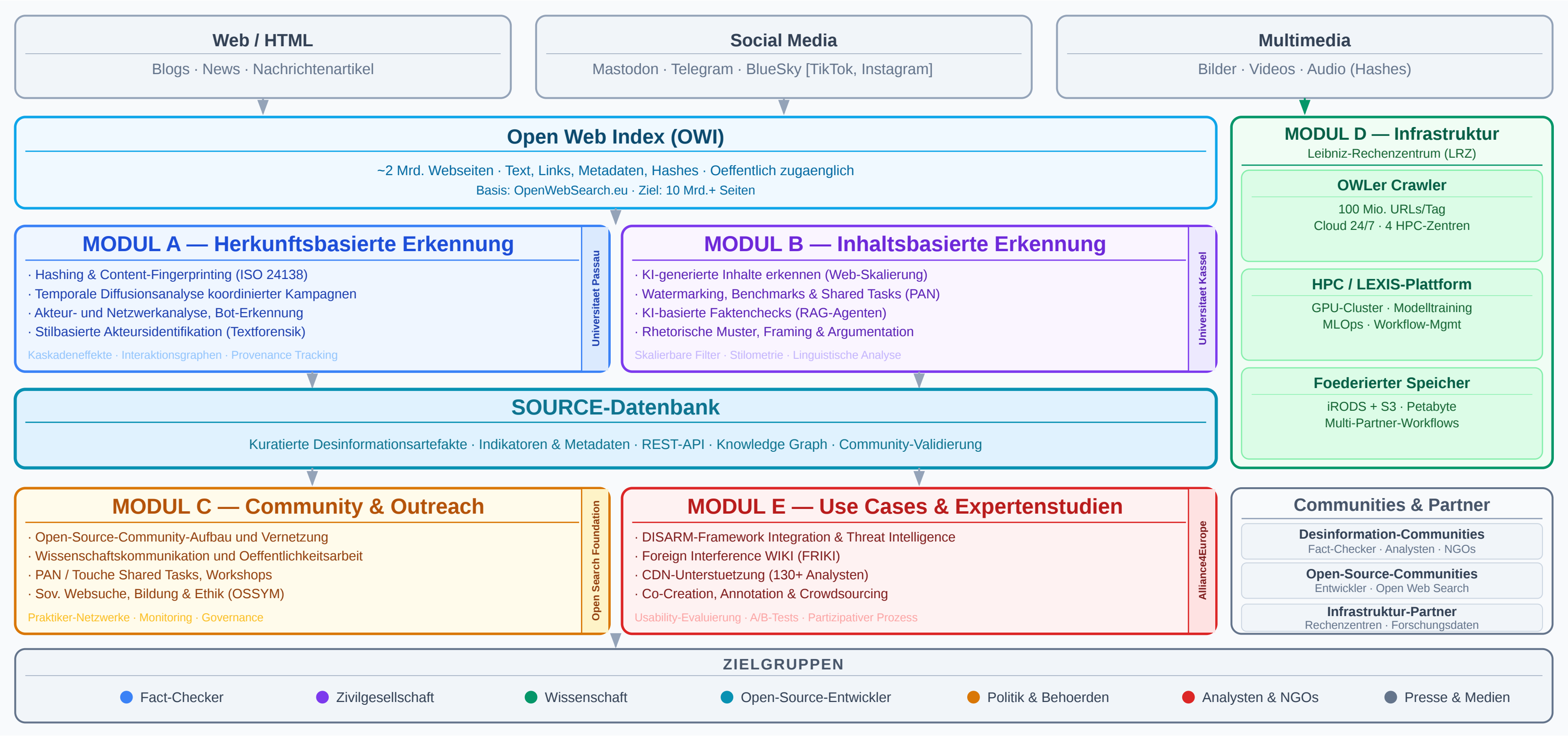
Alliance4Europe

Use cases, analyst workflows, co-creation, and evaluation with practice partners.

Associated communities

Fact-checking, research, and policy-facing communities help validate and operationalise the project outputs.

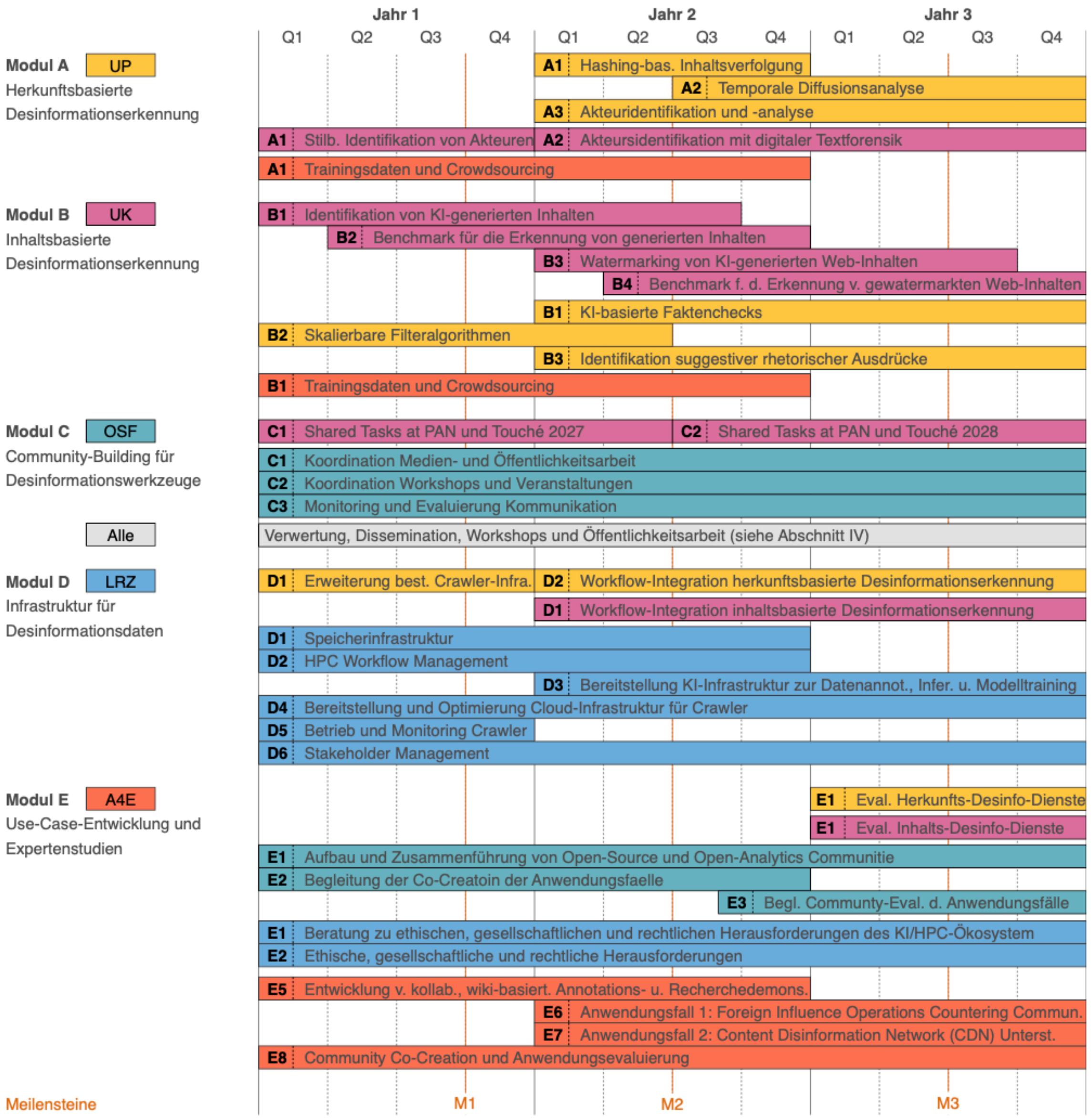
Modules



- **Collect** web and social-media traces at scale
- **Enrich** artefacts with content signals and provenance metadata
- **Link** related content via similarity, timing, actors, and diffusion patterns
- **Expose** results through open interfaces, data services, and analyst-facing tools

SOURCE combines content-based and provenance-based evidence. The project is not just about detecting false claims, but about tracing coordinated campaigns.

Project Timeline



Proposal Milestones

M1 · End of year 1

Core setup completed: actor/content identification started, training data and crowdsourcing in place, crawler and storage foundations established.

M2 · Mid year 2

First integrated workflows: provenance analysis, content analysis, shared-task/community work, and OWI/SOURCE infrastructure integration converge.

M3 · Mid year 3

Evaluated services and use cases: origin/content disinformation services, community evaluation, and operational application scenarios.

2. Background

OpenWebSearch.eu

> Building an Open Index of the Web and an open infrastructure to reduce upfront costs for researchers, innovators and companies in tapping the Web as a resource for search, web analytics, and AI.

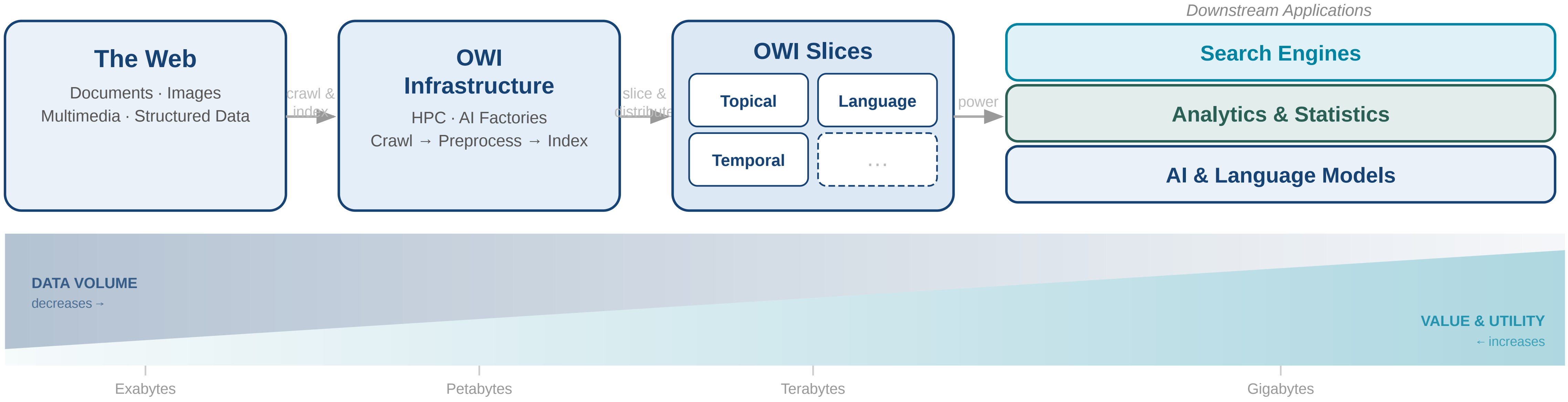
What OpenWebSearch.eu Delivered

- open-source pipelines for crawling, preprocessing, indexing, and distribution
- a federated infrastructure across European HPC and data centres
- dashboard, tools, data access APIs, and public datasets
- an ecosystem for research, RAG, vertical search, and analytics

Why It Matters For SOURCE

- SOURCE can build on an operating web-index backbone
- provenance tracing needs broad web coverage and repeatable access
- open data and transparent workflows reduce dependency on closed search engines
- SOURCE adds disinformation-specific analysis on top

The Open Web Index



The OWI provides:

- **Pull** daily pre-built index shards Daily OWI outputs combine `index.ciff.gz` sparse inverted indices, Parquet metadata and enrichments, and selected dense embeddings for AI-based search and classification.
- **Build** local indices from own data
- **Push** custom indices back into the ecosystem

OWI At Meaningful Scale

Crawling And Indexing

Pages crawled	~39 B total
Unique pages	~12 B
Documents indexed	~10.7 B
Unique domains	~75 M

Access Layer

- **500+** public daily snapshots
- CIFF + Parquet as reusable data formats
- dashboard for browsing datasets and crawl state
- **OUR²S** as hosted search UI and API over OWI content

> For SOURCE, this turns OpenWebSearch.eu from prior work into the technical substrate for large-scale collection, search, verification, and campaign reconstruction.

3. Plans & Goals University of Passau

University Of Passau As Partner

Role in SOURCE

- project coordination
- extension of data acquisition and crawling workflows
- provenance-oriented analysis of content spread
- temporal diffusion modelling and actor-network reconstruction
- integration into shared HPC and storage infrastructure

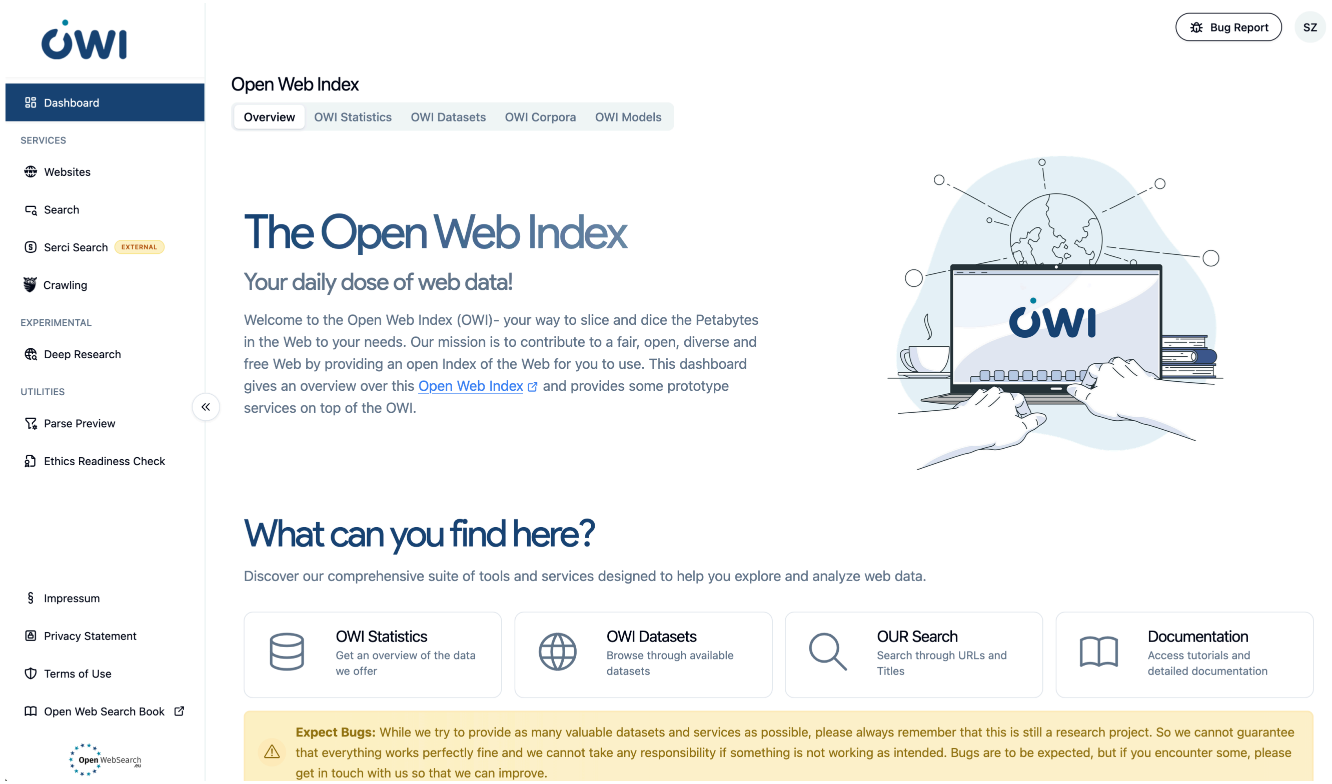
Prior Work In OpenWebSearch.eu

Open Web Index

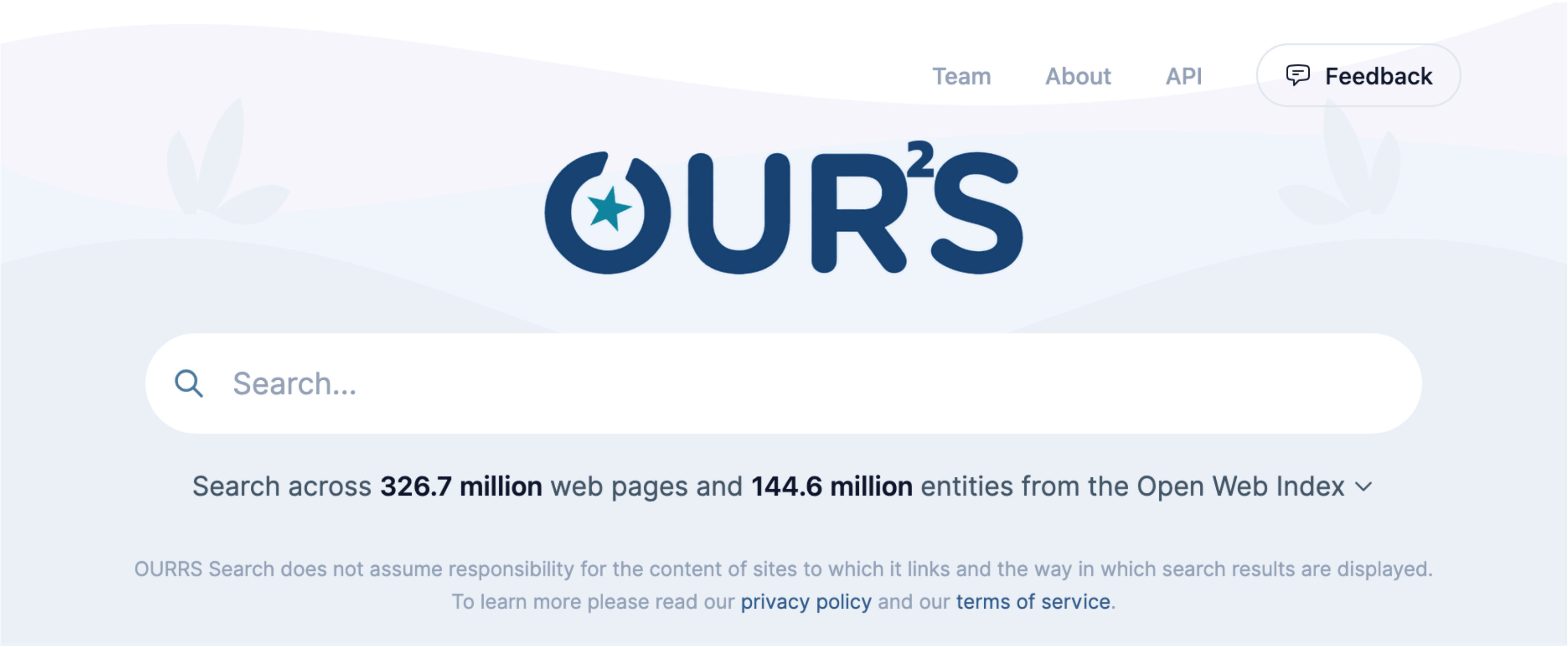
- about **10+ billion indexed documents**
- **500+ public datasets**
- open crawl, index, corpora, and dashboard infrastructure

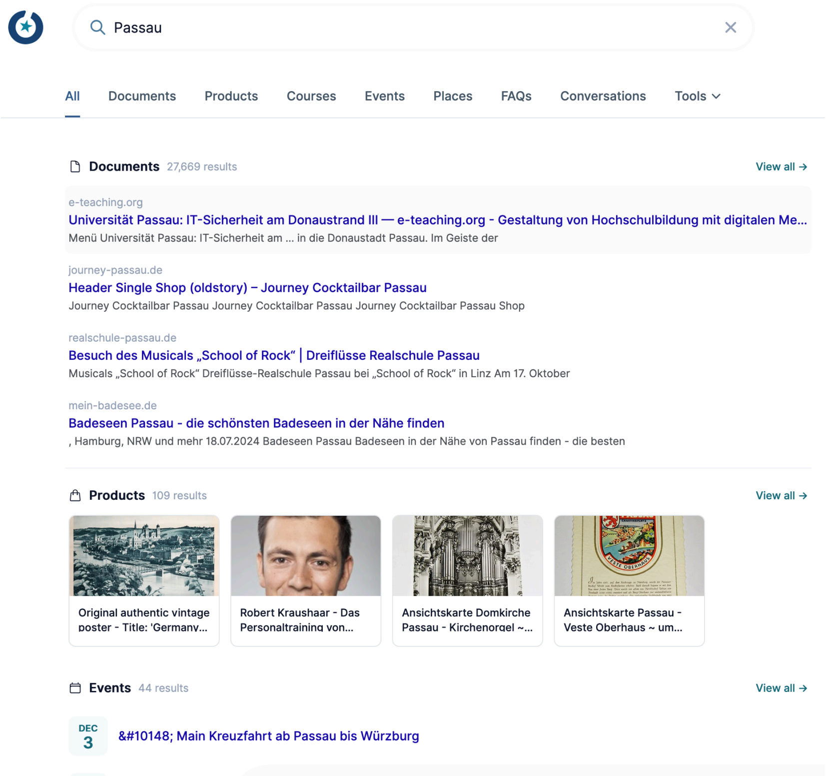
Our Focus in OWSEU

- Crawling & Tooling
- Storage efficient Embeddings
- An API for (Agentic) Retrieval of the OWI



Prior Work: OURRS And Open Access





- **OpenWebSearch.eu** provides the open data and indexing layer.
- **OUR²S** provides a usable retrieval and API layer. Hosts access to OWI content via the Open Web Index Research Retrieval System `ourrs.eu` (Test-Version; Slow & inaccurate)
- **YOARS** Your OpenWebIndex Agentic Research Sytem - an Agent Layer over OUR²S
- **SOURCE** can build on both and add provenance-aware, disinformation-focused analysis services.

4. Goals Uni Passau

Goals From The Proposal

Crawling extension

Expand the existing OpenWebSearch setup toward relevant social-media and campaign-oriented sources.

Similarity tracking

Develop scalable similarity-preserving hashing and chunking to link reused and modified content.

Provenance analysis

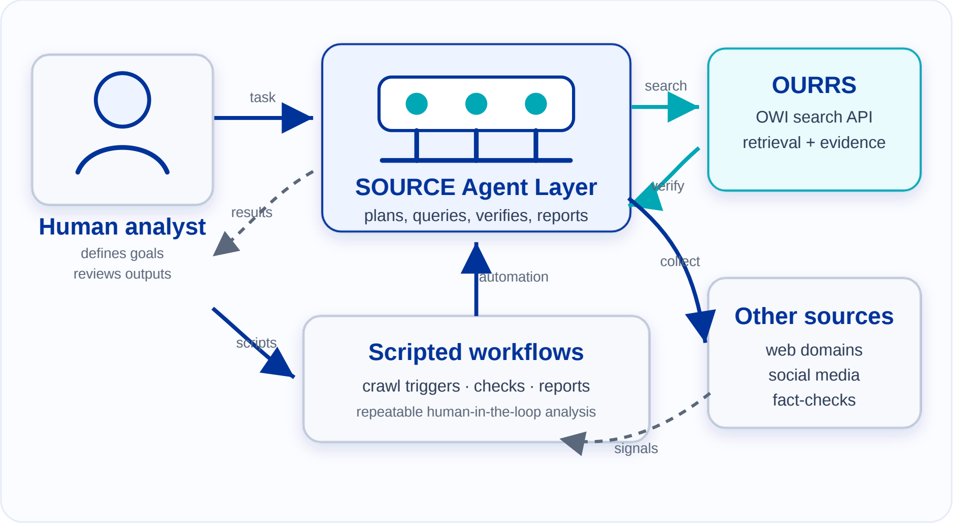
Model temporal diffusion, origin traces, coordination patterns, and actor networks behind campaigns.

Infrastructure integration

Operationalise the methods in shared HPC workflows, storage systems, and interoperable interfaces.

Uni Passau Focus In SOURCE

- extend **OWS** from web indexing to web-plus-social-media monitoring
- connect artefacts through **text similarity**, **image similarity**, and **diffusion signals**
- shift from single-item analysis to **campaign reconstruction**
- support open, reproducible, and reusable workflows for other partners



> The proposal already defines crawling, provenance, and infrastructure work. A practical extension is to add an explicit **agent layer** that automates parts of the disinformation analysis process.

Additional Goal: Agent Layer

Collection agents

Monitor sources, trigger focused crawls, and collect supporting evidence around suspicious artefacts or narratives.

Analysis agents

Summarise provenance signals, compare near-duplicate text and images, and prepare candidate explanations for analysts.

Verification agents

Cross-check claims against known sources, prior items, and campaign patterns to prioritise artefacts for human review.

Workflow agents

Orchestrate pipelines across retrieval, hashing, graph construction, and reporting to reduce analyst overhead.

> This agent layer fits naturally on top of OWS + OURRS + SOURCE infrastructure and turns the platform into an active analysis environment.

5. Next Steps

Next Step 1: Crawling Data Sources

Candidate sources

- Telegram
- Bluesky
- Instagram
- campaign-related web domains and media mirrors
- cross-platform link targets and repost chains

Questions

- Which sources are technically feasible and legally robust?
- Where do we have the highest value for provenance reconstruction?
- Which data can be collected continuously, not only ad hoc?
- How do we align source collection with OWS infrastructure and storage?

Next Step 2: Image Hashes For Provenance Tracking

- create similarity-preserving hashes for images and image variants
- detect reposts, crops, overlays, and small manipulations
- connect images across platforms and across web pages
- use image-level links as additional provenance evidence in campaign graphs

> Image hashing complements text-based provenance and is especially important for memes, screenshots, and reused visual assets in coordinated campaigns.

Next Step 3: Agentic Disinformation Workflow

Pipeline

1. detect suspicious content or narratives
2. expand evidence through search and crawling
3. link text and images via similarity and provenance
4. reconstruct diffusion and actor patterns
5. prepare analyst-facing reports and prioritised alerts

Outcome

- faster triage
- better cross-platform evidence collection
- reproducible analysis chains
- human-in-the-loop validation instead of black-box automation

> The target is not a fully automatic truth machine, but an open workflow that helps analysts trace campaigns more efficiently and transparently.

6. Demo

Summary

By utilizing the Open Web Index, SOURCE can provide a central infrastructure for **large-scale, AI-supported disinformation tracing** and reveal **disinformatino campaigns**.

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Next Steps

Administrativ

- Press Release / Dissemination
- Homepage
- **Kooperationsvertrag**

Roadmap

- Data Sources & Modalities - What content is relevant?
- Use Cases - What to solve?
- Workflows - How to support the community?

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